Kubernetes NetworkPolicy Demo (From Scratch)

Objective

```
Demonstrate how Kubernetes NetworkPolicy controls pod-to-pod communication:
1. Allow communication by default
2. Block traffic with a deny policy
3. Re-allow only specific traffic
```

Architecture

```
frontend pod  --->  backend pod
```

Step 1: Create Namespace

```
kubectl create namespace net-demo
```

Step 2: Backend Pod YAML

```
apiVersion: v1
kind: Pod
metadata:
  name: backend
  namespace: net-demo
  labels:
    app: backend
spec:
  containers:
    - name: nginx
      image: nginx
```

Step 3: Frontend Pod YAML

```
apiVersion: v1
kind: Pod
metadata:
  name: frontend
  namespace: net-demo
  labels:
    app: frontend
spec:
  containers:
    - name: busybox
      image: busybox
      command: ["sleep","3600"]
```

Step 4: Service YAML

```
apiVersion: v1
kind: Service
metadata:
  name: backend-service
  namespace: net-demo
spec:
  selector:
    app: backend
  ports:
    - port: 80
```

Step 5: Test Connectivity

```
kubectl exec -it -n net-demo frontend -- sh
wget -qO- backend-service
```

Step 6: Deny Policy

```
apiVersion: networking.k8s.io/v1
kind: NetworkPolicy
metadata:
  name: deny-all-backend
  namespace: net-demo
spec:
  podSelector:
    matchLabels:
      app: backend
  policyTypes:
    - Ingress
```

Step 7: Allow Only Frontend

```
apiVersion: networking.k8s.io/v1
kind: NetworkPolicy
metadata:
  name: allow-frontend
  namespace: net-demo
spec:
  podSelector:
    matchLabels:
      app: backend
  ingress:
    - from:
        - podSelector:
            matchLabels:
              app: frontend
  policyTypes:
    - Ingress
```

Important Note

```
Use a CNI plugin that supports NetworkPolicy, such as Calico:
minikube start --cni=calico
```